Intermediate elective

Audrey Juneman

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

#colors
library(NatParksPalettes)
```

Data

```r
veg <- read_csv(here("data", "veg.csv"))

metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# create a new object from veg
pools_implicitNA <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP")) |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc) |>
  # calculate mean percent cover
  summarize(mean_pc = mean(percent_cover, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata by year_pool
  left_join(metadata, by = "year_pool")

# create a new object from veg
pools_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP")) |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
           # fill values of percent cover with 0
           fill = list(percent_cover = 0)
           ) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats
  mutate(mean_pc = sum_pc/num_quad)
```

```r
# creating new ojbect from implicit NA data frame
filtered_implicitNA <- pools_implicitNA |>
  # filter to only include
  filter(psoc == "Eryngium_vaseyi")

filtered_explicit0 <- pools_explicit0 |>
  filter(psoc == "Eryngium_vaseyi")

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
                                "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
```

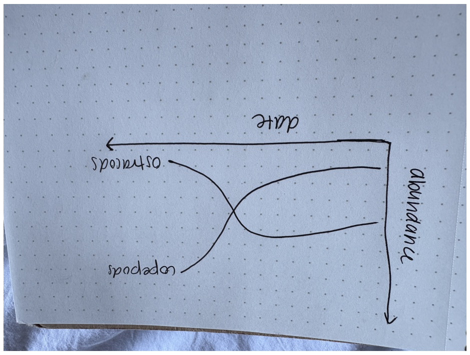
```r
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter
  # (sum abundance divided by 7.5 liters)
```

```
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal,
                                 .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

1. Explain your inspiration

- For my Intermediate elective, I chose the coal falls and renewables rise visualization, due to my interest in energy usage in the United States. I enjoy the color scheme and how easy to interpret the graph is due to the nice colors and organization.
- I think this makes sense for comparing ostracods and copepod abundance throughout time, because it compares two specific variables and shows their trends throughout time. I will specificlaly be looking at Devereaux creek.
- My x axis will be `date_site` and my y-axis will be `taxon`. It will be colored by `taxon`.

2. Plan your figure



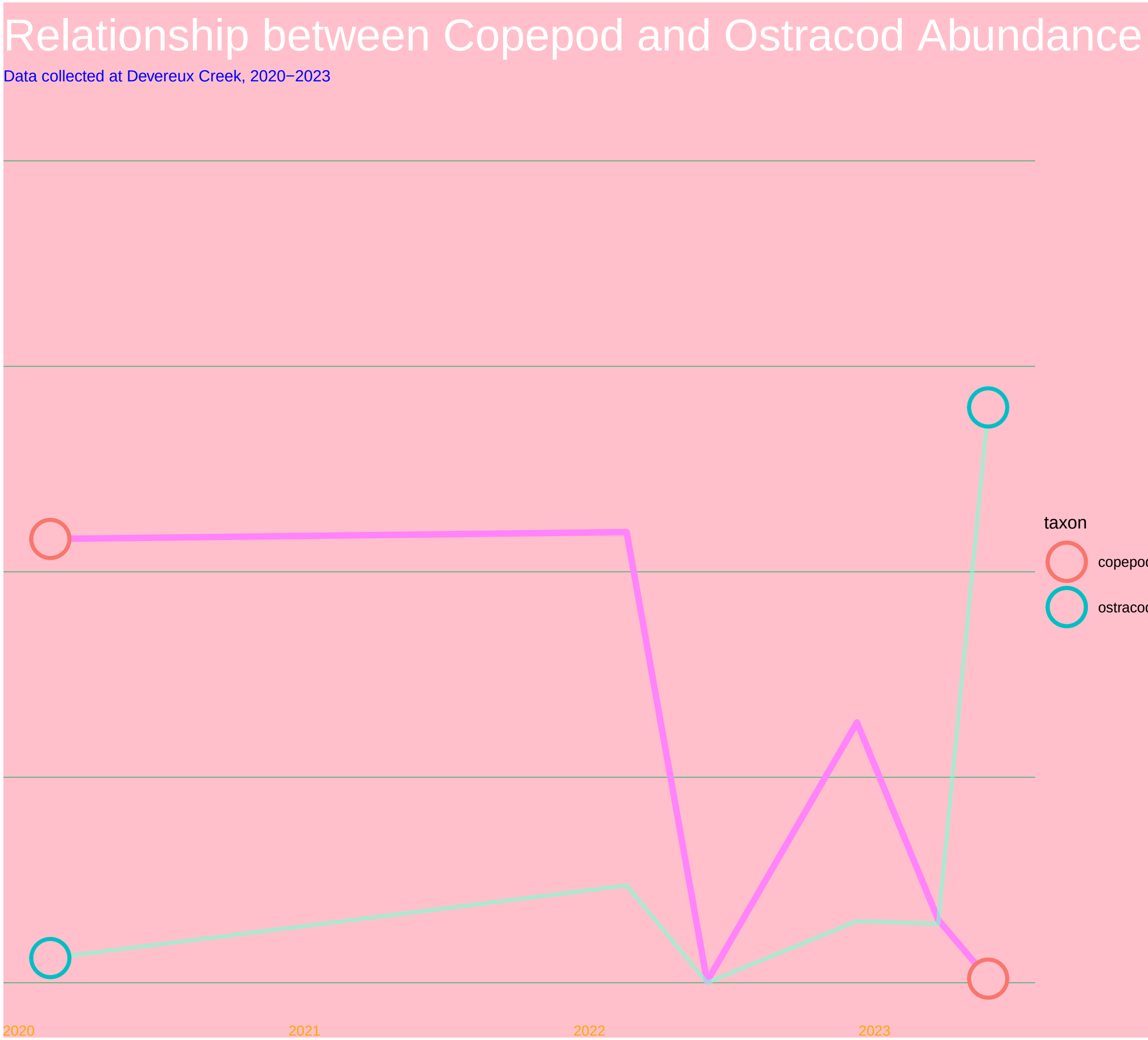
3. Code your figure

```
#filtering dataset for copepod and ostracod line
aq_ins_dev <- aq_ins_clean |>
  filter(site_full == "Devereux Creek") |>
  filter(taxon %in% c("ostracod", "copepod"))
```

```
#| label: plot
#| warning: false

#creating graph
p <- aq_ins_dev |>
  ggplot(aes(x = date, y = ave_lit, color = taxon)) +

  #creating y intercept lines
  geom_hline(
    yintercept = seq(0, 80, by = 20),
    color = "mediumseagreen",
    linewidth = 0.25
  ) +
  # creating copepod line
  geom_line(
    data = aq_ins_dev |> filter(taxon == "copepod"),
    linewidth = 1.9,
    color = "orchid1",
    alpha = 1.0
  ) +
```

```r
#creating ostracod line
geom_line(
  data = aq_ins_dev |> filter(taxon == "ostracod"),
  linewidth = 1.3,
  color = "aquamarine",
  alpha = 0.60
) +
#adding point colors
geom_point(
  data = aq_ins_dev |>  filter(date == as_date("2020-01-21")),
  size = 10,
  shape = 21,
  fill = "pink",
  stroke = 1.8
) +
#adding point colors
geom_point(
  data = aq_ins_dev |> filter(date == as_date("2023-05-06")),
  size = 10,
  shape = 21,
  fill = "pink",
  stroke = 1.8
) +
# adding labels for axes
labs(
  title = "Relationship between Copepod and Ostracod Abundance",
  subtitle = "Data collected at Devereux Creek, 2020-2023",
  x = "Date",
  y = "Abundance"
) +
theme_void() +
theme(
  plot.background = element_rect(fill = "pink", color = NA),
  panel.background = element_rect(fill = "pink", color = NA),
  plot.title = element_text(
    size = 28,
    color = "white",
    hjust = 0,
    lineheight = 1.1,
    margin = margin(t=10, b=6)),
  plot.subtitle = element_text(
    size = 10,
```

```
      color = "blue",
      hjust = 0,
      lineheight = 1.4,
      margin = margin(b=22)),
    plot.caption = element_text(
      size = 6.5,
      color = "red",
      hjust = 1,
      margin = margin(t = 14)),
    axis.text.x = element_text(
      size = 9,
      color = "orange",
      hjust = 1
    )
  )

p
```



4. Describe your process

- This process was extremely diffucult and I did not feel as though I had the tools to use the other persons code, just because of how complicated certain aspects of the code were. It felt extremely stressful, and I wish we could have practiced this more in class. I kept having errors I did not understand, and ended up having to remove most of the code from the website. This visualization has custom colors, and it compared one specific site throughout time.